

Stat 3113: Lesson 9

Independence

Independence

Definition: Two events are **independent** if $P(A | B) = P(A)$, and **dependent** otherwise.

Example: Toss a fair 6-sided die once. *→ Equally Likely Outcomes.*

Let $A = \{2, 4, 6\}$, $B = \{1, 2, 3\}$, and $C = \{1, 2, 3, 4\}$.

Notice,

$$P(A) = \frac{3}{6} = \frac{1}{2}$$

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{P(\{2\})}{P(\{1, 2, 3\})} = \frac{1/6}{3/6} = \frac{1}{3}$$

$$P(A | C) = \frac{P(A \cap C)}{P(C)} = \frac{P(\{2, 4\})}{P(\{1, 2, 3, 4\})} = \frac{2/6}{4/6} = \frac{1}{2}$$

So, A and B are **dependent**, but A and C are **independent**.

Note: Independence and **mutually exclusive** are **NOT** the same.

Independence of Other Events

If the events A and B are **independent**, the following events are also independent:

1. A' and B
 2. B' and A
 3. A' and B'
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Multiplication Rule for Independence

- When A and B are independent, $P(A \cap B) = P(A) \cdot P(B)$.

Similarly,

- When A , B , and C are independent, $P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$

Etc.

So, when events are independent, the probability of their intersection is the product of their probabilities.

Example: Suppose that the proportions of blood phenotypes in a particular population are as follows:

A	B	AB	O	<u>Total</u> = 1
0.4	0.11	0.04	0.45	

Assuming that the phenotypes of two randomly selected individuals are independent of one another,

a. what is the probability that both phenotypes are O?

$$IP(O_1 \cap O_2) = IP(O_1) \times IP(O_2) = 0.45 \times 0.45 = 0.2025$$

↑
(by independence)

b. what is the probability that the phenotypes of two randomly selected individuals match?

$$\begin{aligned}
 IP(\text{"Phenotypes match"}) &= IP(O_1 \cap O_2) + IP(A_1 \cap A_2) + IP(B_1 \cap B_2) + IP(AB_1 \cap AB_2) \\
 &= IP(O_1) \times IP(O_2) + IP(A_1) \times IP(A_2) + IP(B_1) \times IP(B_2) + IP(AB_1) \times IP(AB_2) \\
 &= 0.45^2 + 0.4^2 + 0.11^2 + 0.04^2 = 0.3762
 \end{aligned}$$

Reading

Chapter 3 – Section 3.1

Next Topic



Random Variables



Chapter 3 - Section 3.3